SYNTHETIC RANDOMIZED CLINICAL TRIAL · N = 640

ctDNA-Guided Dual Checkpoint Blockade Reduces 12-Month Recurrence in Stage III Melanoma

POPULATION & SCREENING

N = 640 Adults

- Resected stage IIIB–IIIC melanoma
- Post-op ctDNA positive within 6 weeks
- 18 academic oncology centers

RANDOMIZED REGIMEN

1:1 Allocation (12 Cycles)

- Dual PD-1 + LAG-3 blockade (n = 320)
- Standard PD-1 monotherapy (n = 320)
- Serial ctDNA surveillance every 6 weeks

12-MONTH OUTCOMES

HR 0.58 (95% CI 0.43–0.79)

- Week-12 ctDNA clearance: 68% vs 44%
- Grade >=3 irAEs: 21.8% vs 14.6%

SYNTHETIC EXAMPLE DATA FOR SKILL VALIDATION · Primary Endpoint HR 0.58 (95% CI 0.43–0.79; P = 0.0004)